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Vibration reduction in the cabins of heavy-duty trucks using the theory of load transfer paths

Hiroaki Hoshino^a, Toshiaki Sakurai^b, Kunihiro Takahashi^c

^a *Mechanical Dynamics Japan K.K., 5-2-39 Akasaka Minato-ku, Tokyo 107-0052, Japan*

^b *Graduate School of Keio University, 3-14-1 Hiyoshi Kohoku-ku, Yokohama, Kanagawa 223-8522, Japan*

^c *Keio University, 3-14-1 Hiyoshi Kohoku-ku, Yokohama, Kanagawa 223-8522, Japan*

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Abstract

The objective of this study is to investigate the load transfer paths in the cabin structures of heavy-duty trucks under static loading, and to apply the results to reduce vibration in cabins. In a preliminary simulation using a simple model, it is shown that the floor panel vibration is closely related to the stiffness of the front cross-member of the floor structure. Load path U^* analyses using the finite element method show that the low stiffness of the front cross-member is caused by discontinuities and nonuniformities in the load paths. © 2003 Society of Automotive Engineers of Japan, Inc. and Elsevier Science B.V. All rights reserved.

1. Introduction

The reduction of noise in motor vehicle compartments has become increasingly important in recent years. This trend now includes the heavy-duty truck. The booming noise in a truck cabin has been widely studied [1–3] to improve cabin quietness. In general, noise and vibration can be effectively reduced by stiffening the rigidity of a body structure. However, it is necessary to investigate truck cabs in detail to realize high-rigidity bodies that are compatible with lightweight and high-crashworthiness structures.

In our previous report [4], a method was introduced to estimate the influence of floor panel vibration upon the booming noise in a heavy-duty truck using a finite element model. A robust design method called the Taguchi method was also applied in the previous paper, and it was found that increasing the plate thickness of the front cross-member was most effective for reducing the booming noise. This result shows that the reinforcement for a supporting frame member is more effective than that for the panel itself to reduce the vibration of the panel.

In the next section, for a preliminary study, the relation between the values of the spring constants for connecting components and the characteristics of cab floor vibration is discussed. A simple mass-spring model is used for this analysis.

In strict investigations into structural vibration, it is important to consider the influence of the vibration

modes [5]. However, once the relation between the stiffness of connecting springs and the floor vibration is found in the simple model mentioned above, vibration reduction can be discussed as a static problem of the structure.

Based on this idea, the concept of load path analysis [6] is introduced to evaluate the static problem regarding the cab floor panel. This new method is used to find the load paths from the cab mount to the cab structure and to discuss the role of each structural component.

The objectives of this study are to investigate the static load transfer paths in the cab floor panel and to point out the structural discontinuity in the front cross-member that relates strongly to the booming noise as suggested in the previous report.

2. Simple model

A simple mass-spring model for the truck cab is developed to relate the vibration problem to the static structural analysis.

2.1. Simple cab model

As shown in Fig. 1, a cab is divided into five principal rigid masses in the mass-spring model: the cab mount, the side-member, the front cross-member, the floor

panel and the upper body. These masses are connected to each other with linear spring elements.

Spring constants are calculated by using the FE model shown in Section 4 in the static condition.

2.2. Frequency response analysis

The eigenvalues and the vibration levels of the floor panel in Fig. 1 are calculated under the condition of excitation at the cab mount using ADAMS/Vibration.

The result of the frequency response analysis is shown by the thin line in Fig. 2, which can be compared with the experimental results shown by the bold line in Fig. 2. Since the vibration mode of each panel is not accounted for in this simple model, it should be noted that the results are not strictly accurate below the lowest natural frequency. The lowest natural frequency can be observed at 118 Hz in Fig. 2. However, the vibration modes with the frequency below 100 Hz are not the modes of the floor panel and we focus on the

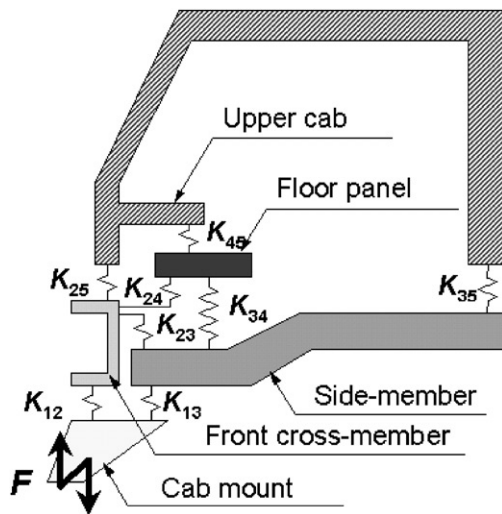


Fig. 1. Simple cab model for vibration analyses.

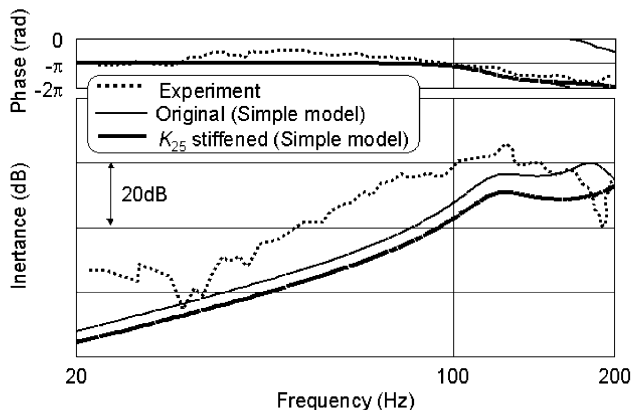


Fig. 2. Results of the experiment and calculations.

100–200 Hz frequency region to investigate the frequency characteristics and magnitude of inertance. For these reasons, it can be said that the simple model represents the dynamic characteristics within this frequency region.

Based on the results of the previous report [4], that the reinforcement of the cross-member effectively reduces floor panel vibration, the frequency response is calculated for the simple model in which the value of the spring constant K_{25} is double the original value. The result of this calculation is shown in Fig. 2, indicating the effectiveness of the value of K_{25} , which corresponds to the above expectation.

From these calculations, it is shown that the vibration problem of the floor panel can be replaced by the static structural problem of a cab structure.

3. Load transfer path for static analysis

In this section, the concept of load transfer path in a static analysis is summarized briefly.

3.1. Concept of load path analysis [6–8]

3.1.1. Connectivity index

Fig. 3 shows a structure that has a loading point A and a support point B. The position of an arbitrary point C is also indicated in the structure. When point C is unrestrained or free, the strain energy stored in the whole body is denoted by U shown in Eq. (1), where p_A and d_A indicate the vector values of a loading and a displacement, respectively. When point C is fixed, the strain energy to give the same displacement is expressed by U' as shown in Eq. (2), where p'_A is a new loading value. In Eq. (2), it should be noted that we use the same value d_A for the displacement as the value of the displacement in Eq. (1).

$$U = \frac{1}{2} p_A \cdot d_A, \quad (1)$$

$$U' = \frac{1}{2} p'_A \cdot d_A. \quad (2)$$

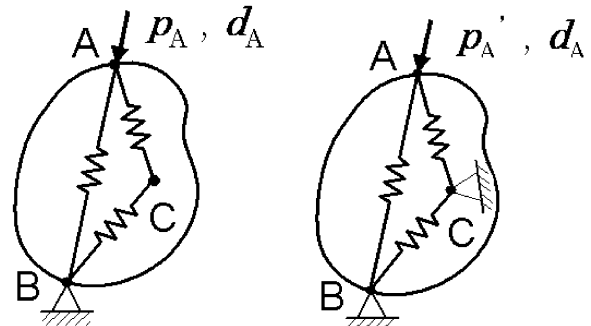


Fig. 3. Concept of internal stiffness.

Using the values of U and U' , we introduced a non-dimensional value U^* , shown in Eq. (3) as an index of connectivity, that indicates the strength of connectivity between the loaded point A and an arbitrary point C in the structure.

$$U^* = 1 - \frac{1}{U/U'}. \quad (3)$$

3.1.2. Load transfer path

Fig. 4 shows a schematic diagram of the contour lines of U^* . The curve from A to B along the ridgeline of the contour plot is defined as the load transfer path.

3.1.3. Conditions for a desirable structure

Conditions for a structure with a desirable load path are demonstrated schematically in Fig. 5, where s is the distance along a path from the loading point A and l is the whole length of a path. Three conditions are summarized as follows:

Uniformity: minimization of the shaded area in Fig. 5(a)

Continuity: minimization of the shaded area in Fig. 5(b)

Path consistency: minimization of the shaded area in Fig. 5(c), where S_1 and S_2 are the load paths from A to B and B to A, respectively.

3.2. Optimization using U^* and the three conditions [8]

A result of the optimization calculation for the simple structure shown in Fig. 6(a) was previously reported [8] by the authors.

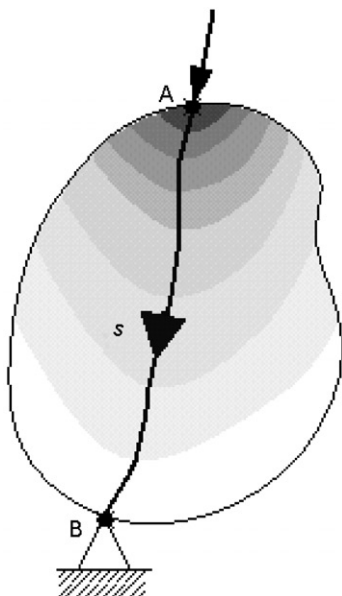


Fig. 4. Load transfer path.

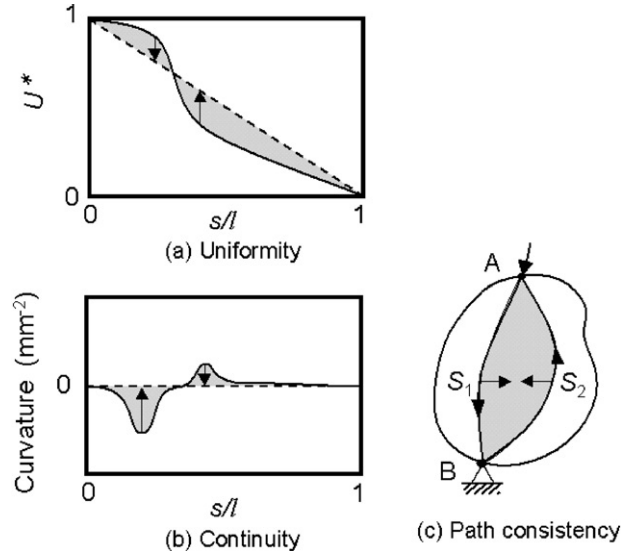


Fig. 5. Three conditions for the desirable structure.

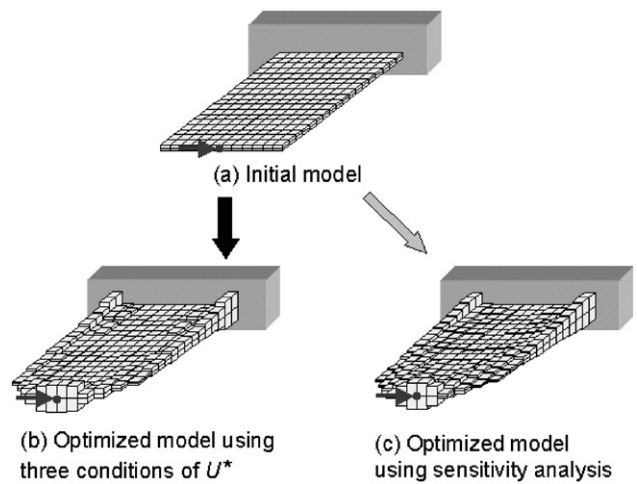


Fig. 6. Optimized structure.

The optimized shape shown in Fig. 6(b) was obtained using the three conditions of U^* as objective functions. The optimized shape in Fig. 6(c) shows the results of the conventional sensitivity analysis minimizing the strain energy density. These two structures have similar thickness distributions. It is noteworthy that the optimization process in Fig. 6(b) never used the concepts of stress, strain or strain energy.

4. Load transfer in cab floor structure

It was found that the stiffness of the connecting spring K_{25} in Fig. 1 is important, based on both the results with the simple model and those of the previous report [4]. To explain the reasons for these results, in this section, we apply the U^* analysis to a cab floor structure.

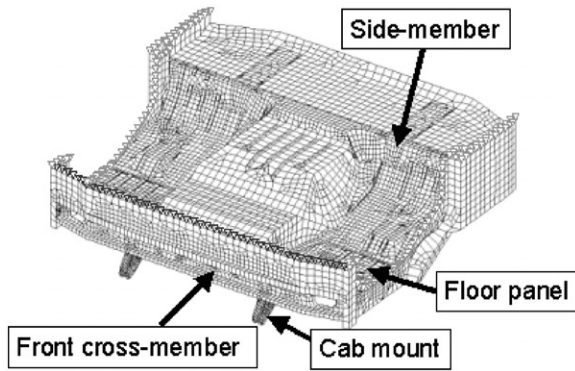


Fig. 7. Cab floor FE model.

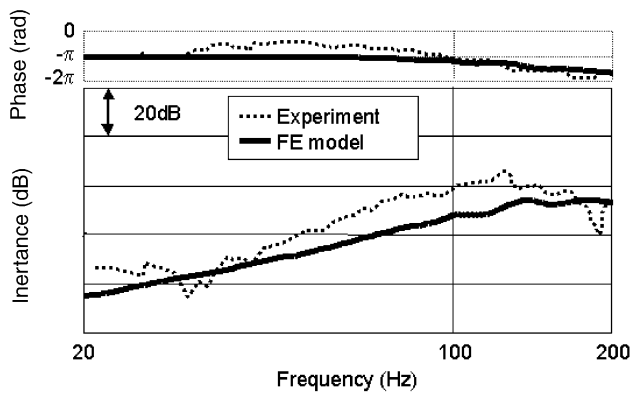


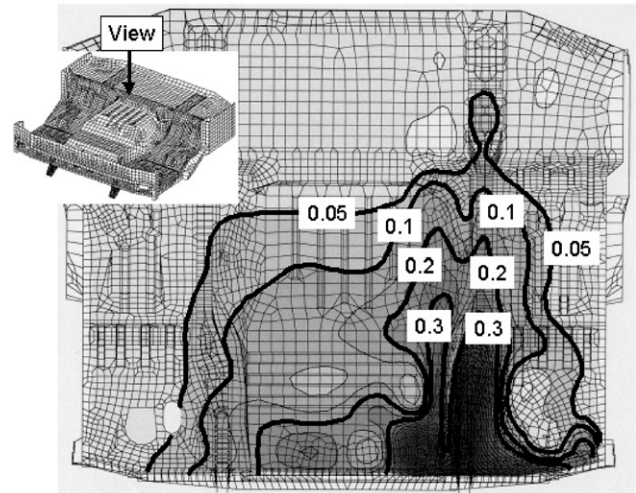
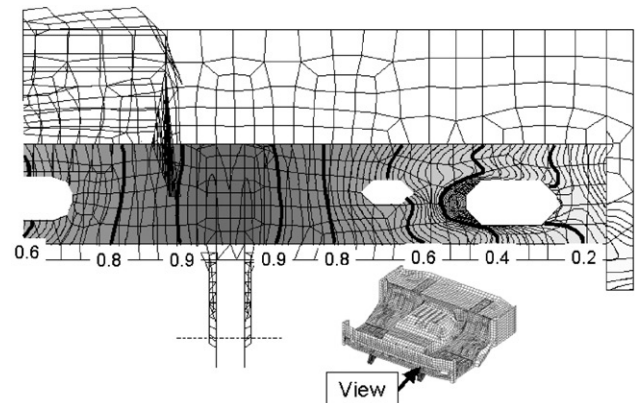
Fig. 8. Results of experiment and FE model.

4.1. Finite element model for U^* analysis

An FE model of a cab floor panel with 10742 nodes and 11493 shell elements is shown in Fig. 7. The vibration level of the left-hand side floor panel is calculated under the condition of excitation at the left cab mount using NASTRAN.

The inertance of the FE analysis is shown in Fig. 8 for comparison with the test results. We can see that there is no strict correspondence between the tests and the simulations. This disagreement is attributed to the local mass distribution not being strictly reproduced in the FE model.

On the other hand, the plot of inertance for FE model shows a straight line below 100 Hz. This means the stiffness is constant in this frequency region. Further, no resonant mode can be seen below 100 Hz. Therefore, we can say that this stiffness value is one of the factors to determine the lowest resonant mode and is almost equivalent to the stiffness value obtained in a static load analysis. The stiffness characteristic obtained from the calculation corresponds well with that of the experiment in terms of the overall stiffness. For this reason, we think that the above FE model can be used for static analyses to evaluate the floor structure.

Fig. 9. U^* distribution (Floor panel).Fig. 10. U^* distribution (Front cross-member).

4.2. Load transfer path in cab floor

The values of U^* and load transfer paths are calculated under the condition of excitation at the left cab mount using the FE model. Static calculations are performed in shifting the evaluation point on the floor that corresponds to point C in Fig. 3. The value of U^* at each point is calculated by substituting the results in Eq. (3). The results of U^* are shown in Figs. 9 and 10. Three paths are drawn from the U^* distribution automatically by post-process calculations (Figs. 11–13).

4.3. Evaluation of conditions for desirable structures

The uniformity and continuity in the path analysis are evaluated with regard to the three load transfer paths.

- (1) Path 1. The conditions of uniformity and continuity of path 1 are expressed in Fig. 14. For both conditions, remarkable changes are observed at

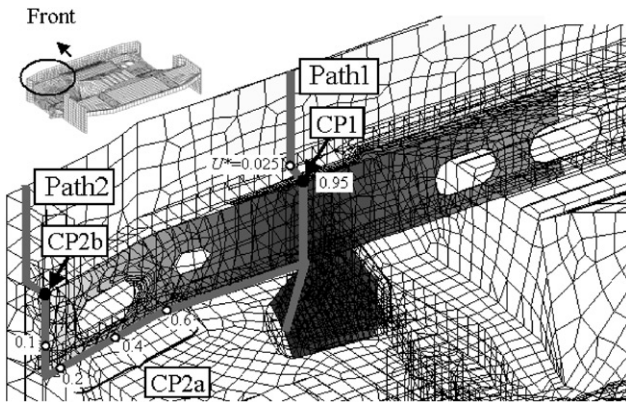


Fig. 11. Load transfer paths (Path 1 and Path 2).

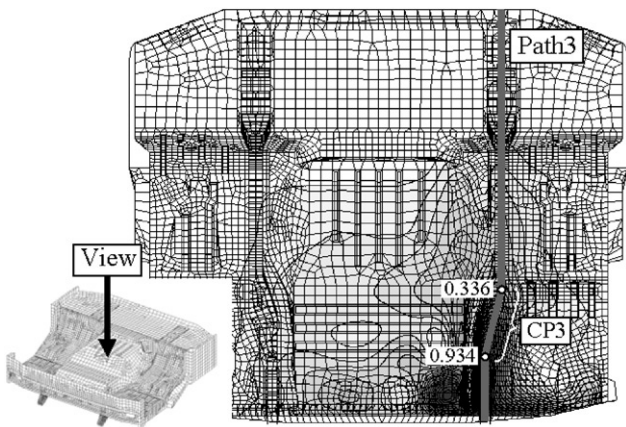


Fig. 12. Load transfer path (Path 3).

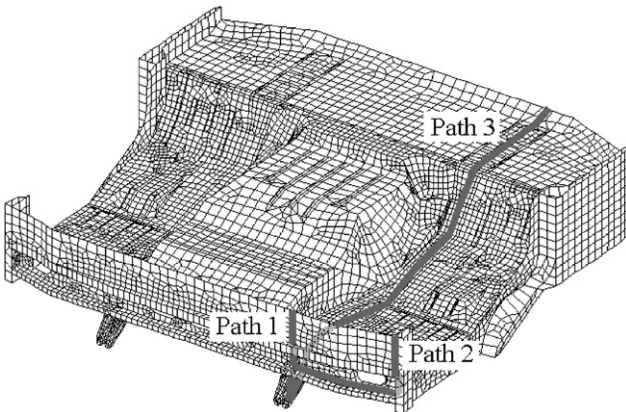


Fig. 13. Load transfer paths (overview).

the position where the front cross-member and the front wall are connected. The critical point of U^* is denoted by CP1 in Figs. 11 and 14. It is desirable to increase the stiffness between the cross-member and the front wall and to use the front wall effectively as a shear panel. It is also suggested that the cross-

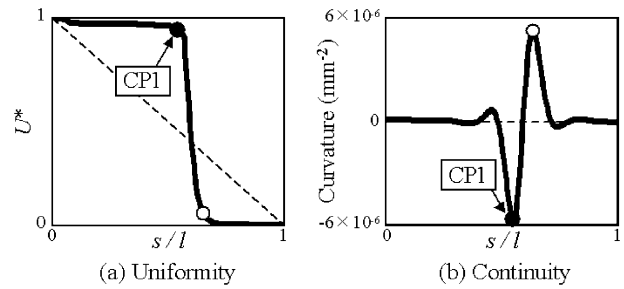


Fig. 14. Uniformity and continuity of Path 1.

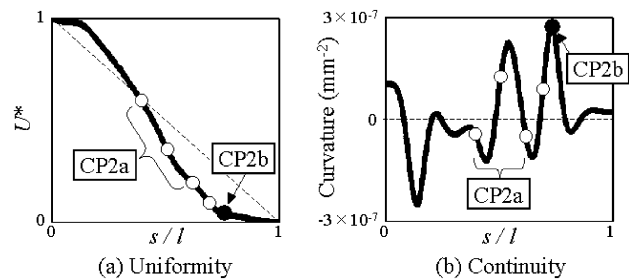


Fig. 15. Uniformity and continuity of Path 2.

sectional shape of the cross-member should be modified to improve the stiffness of the connection between the cross-member and the front wall.

- (2) Path 2. The uniformity and continuity of path 2 are expressed in Fig. 15. Values of U^* along path 2 decrease rapidly due to the elliptic hole denoted by CP2a in Figs. 11 and 15. The shearing force in the web plate of the cross-member plays an important role in transferring the input load to the lower part of a pillar. The elliptic hole is located at the center of this main transfer path. It is also observed that the load transfer is unstable at the connection between the lower part of the pillar and the cross-member denoted by CP2b in Figs. 11 and 15. The stiffness of the connection at this position is to be increased.
- (3) Path 3. The expressions of uniformity and continuity of path 3 are shown in Fig. 16. From the standpoint of uniformity, even though a member transfers the load by bending along path 3, input loading is transferred desirably. Regarding continuity, internal stiffness is rather unstable at a bend in the side-member, which is denoted by CP3 in Figs. 12 and 16.

From the above results, it can be said that Path 1 and Path 2 have some nonuniformities and discontinuities in the front cross-member that reduce the stiffness of the member. This observation corresponds well with the result obtained in the previous paper [4].

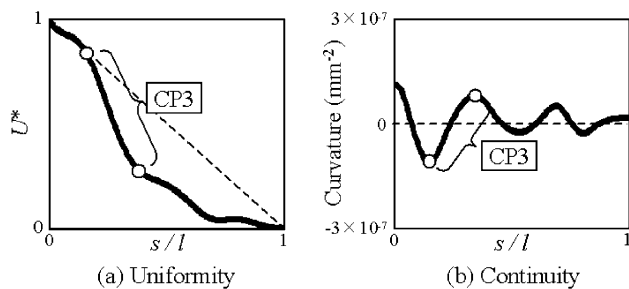


Fig. 16. Uniformity and continuity of Path 3.

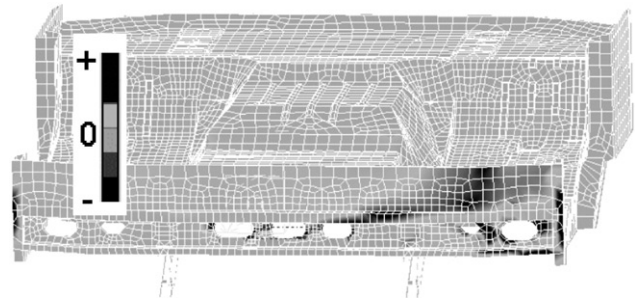


Fig. 17. Maximum principal stress distribution.

4.4. Evaluation using stress distribution

Fig. 17 shows the distribution of maximum principal stresses under the condition of vertical static loading at the left-hand cab mount. Comparing the stress distribution and the U^* distribution in Fig. 10, we can observe some discrepancies. Although a prominent stress concentration is seen around the elliptic hole in Fig. 17, it is not correct to conclude that a hole is effective to transfer loads. Therefore, we can say that the stress value is not always adequate to indicate the value of a transferred load. Even though the stress value can be a most important index of the strength of structures, the value of U^* is more effectively used for the evaluation of transferred loads and overall rigidities.

5. Discussion for cab floor structure

We obtained the load transfer paths in a cab floor structure, and examined the differences between those paths and the most desirable ones.

The reason why the vibration of a floor panel is reduced by increasing the thickness of a front cross-member was not revealed in the previous paper. In this study, the solution is suggested by an investigation of the discontinuities and nonuniformities in paths 1 and 2. The improvement of these paths is necessary to increase the stiffness of K_{25} (in Fig. 1) and to build a balanced cab structure.

6. Conclusions

In this study, the load transfer in a cab structure of a heavy-duty truck was investigated using the load path U^* analysis, and the relationship between the reduction of vibration and the improvement of load transfer paths was discussed.

The obtained results are summarized as follows:

- (1) A simple mass-spring model of a cab structure for a heavy-duty truck is calculated for the frequency response analysis. The calculated results indicate

the effectiveness of increasing the stiffness value of K_{25} in Fig. 1. It is also shown that the vibration problem of the floor panel can be replaced by the static structural problem of a cab structure.

- (2) A finite element model of a cab floor is developed for the load transfer path analysis under a static condition. Three paths are obtained from the U^* distribution. It is shown from the U^* analysis that two paths have some nonuniformities and discontinuities in the front cross-member which cause the low stiffness value of K_{25} .
- (3) In the conventional analysis, stress values are generally used to express load transfer paths. However, since the local effect of stresses, such as a stress concentration, cannot be distinguished from the overall effect, stress is not always an adequate indicator of the transferred load. Values of U^* are more effectively used for the evaluation of transferred loads and overall rigidities.

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